

SMRI working group notes
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Expander Graph : Spectral Geometry on Graph and Cayley graph

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0 Introduction

This group work as per goal to work on expander graph working through mostly the paper of Alexander Lubotzki [1]. On our 3 first sessions, we set some basic notions and results on expander graph and we also looked at some application through probability. I will firstly re-introduce some of these notions and result and then we will see another way to look at expander graph through spectral geometry and we will finish by defining Cayley graph and some fun result. I will firstly state that we will only work on finite graph.

Definition 0.0.1 (Expander graph)**.** Let $X = (V, E)$ be a finite graph and let $\varepsilon > 0$.

We say that X is an ε-expander if, for every subset $Y \subseteq V$ satisfying $|Y| \leq \frac{|V|}{2}$, we have

$$|\partial Y| \geq \varepsilon |Y|,$$

where

$$\partial Y = \{v \in V \setminus Y : \exists\, u \in Y,\ \{u, v\} \in E\}$$

is the vertex boundary of Y.

The largest such ε is denoted by $\varepsilon(X)$.

Definition 0.0.2 (Cheeger constant)**.** Let $X = (V, E)$ be a finite graph. The Cheeger constant of X is defined by

$$h(X) = \min_{\substack{V = Y_1 \sqcup Y_2 \\ Y_1, Y_2 \neq \varnothing}} \frac{|E(Y_1, Y_2)|}{\min\{|Y_1|, |Y_2|\}},$$

where $E(Y_1, Y_2)$ denotes the set of edges joining a vertex of Y_1 to a vertex of Y_2.

Proposition 0.0.3 (Cheeger inequalities)**.** *Let X be a finite k-regular graph. Then*

$$\frac{h(X)}{k} \leq \varepsilon(X) \leq h(X).$$

Definition 0.0.4 (Walk, Trail, Path)**.** Let $G = (V, E)$ be a graph.

- A *walk* in a graph G is a sequence of vertices $(v_0, ..., v_n)$ s.t. $\{v_i, v_{i+1}\} \in E$ (i.e $\{v_i, v_{i+1}\}$ form an edge of E).
- A *trail* is a walk that does not repeat any edges. If the trail starts and ends at the same vertex (i.e. $v_0 = v_n$) then it is called a *circuit*.
- A *path* is a trail that does not repeat any vertices except perhaps for $v_0 = v_n$ and in that cased the path is called a *cycle*.

Example 0.0.5. Let G be the 5 vertices graph below,

- $\{0, 1, 2, 0, 1, 0\}$ is a closed walk but not a trail or a path because it's backtracking $0, 1, 0$.

- $\{0, 1, 2, 0, 4, 3, 0\}$ is a circuit (closed trail) but it's not a path because it passing by 0 3 times.
- $\{0, 1, 2, 0\}$ is a cycle (closed path).

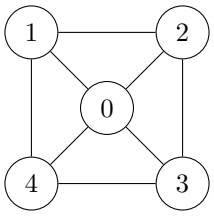
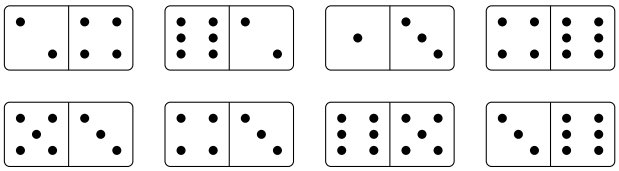


Figure 0.0.1: 5 vertices graph

Question 0.0.6 (Puzzle)**.**



If I match these dominoes to make a line (i.e. we match 2 dominoes if they have the same number on the side we match them), what will be the sum of the number of the 2 ends ?

1 Spectral Geometry

1.1 Adjacency matrix, eigenvalues and Laplacian

Definition 1.1.1 (Adjacency matrix)**.** Let $G = (V, E)$ be a graph with $|V| = n$ The adjacency matrix associated to the graph G is the $n \times n$ matrix such that $A_{i,j}$ is the number of edges from i to j.

Example 1.1.2 (Adjacency matrix of a complete graph)**.** Let G be the complete graph, then $A = \begin{pmatrix} 0 & 1 & \cdots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & 0 \end{pmatrix}$

Remark 1.1.3. For a k-regular graph, the sum of the coefficient of a colone is equal to k.

Remark 1.1.4. A acts on the space of vertex function $\mathbb{R}^V = \{f : V \to \mathbb{R}\}$ such that $A \cdot f = \sum_{y \sim x} f(y)$ where $x \sim y$ means y is a "neighbor" of x i.e. all the vertices connected to x by an edge.

Lemma 1.1.5. *Let $G = (V, E)$ and $G' = (V', E')$ 2 graphs isomorphic then A up to permutation is equal to A'*

Proof. By definition, there exist a bijection $\varphi : V \to V'$ such that:

$$\{u, v\} \in E \iff \{\varphi(u), \varphi(v)\} \in E'$$

the adjacency matrix A is define A_{ij} = number of edges between v_i and v_j in E
If we choose the order of the vertices in G' such that $V' = \{\varphi(v_1), \dots, \varphi(v_n)\}$ then the matrix of adjacency of G' is define as :
A'_{ij} = number of edges between $\varphi(v_i)$ and $\varphi(v_j)$ in E, but then because of the isomorphism $A_{ij} = A'_{ij}$. And if we don't choose this order we just need to make a permutation. □

Definition 1.1.6 (Laplacian)**.** The Laplacian of a graph is the operator s.t. :

$$\Delta f(x) = \sum_{y \sim x} f(y) - f(x) = A \cdot f(x) - kf(x)$$

Hence $\Delta = A - kI$

Proposition 1.1.7. *If $\lambda_1 \geq \cdots \geq \lambda_n$ is the spectrum of the adjacency matrix A associated to a k-regular graph $G = (V, E)$ then we can prove that:*

- $\lambda_1 = k$ *and even more* $\lambda_i \in [-k, k]$
- $\lambda_2 = k \iff G$ *is not connected*
- $\lambda_n = -k \iff G$ *contains a non empty bipartite graph as a connected component*

Proof. Firstly, we easily see that k is an eigenvalue with the vector $\mathbf{1}$: $A \cdot \mathbf{1} = k\mathbf{1}$
And then we can show that this is the highest value an eigenvalue can have by studying any eigenvector $x = (x_1, \dots, x_n)^T \neq 0$ with $A \cdot x = \lambda x$. If we take $|x_i| = \max_{1 \leq j \leq n} (|x_j|)$, then

$$\lambda x_i = \sum_{j=1}^{n} A_{ij} x_j$$

and because $\forall j, |x_j| \leq |x_i|$ and because G is a k-regular graph:

$$|\lambda||x_i| \leq \sum_{j=1}^{n} A_{ij}|x_j| \leq |x_i| \sum_{j=1}^{n} A_{ij} = k|x_i|$$

Hence $\lambda \in [-k, k]$ and $\lambda_1 = k$.

For the second point:
¡= With $X = X_1 \sqcup X_2$ not connected, we just need to use:

$$\left.\begin{array}{l} f_1(i) = \mathbb{1}_{X_1}(i) \\ f_2(i) = \mathbb{1}_{X_2}(i) \end{array}\right\} \text{both eigenvector associated to } k$$

Hence $\lambda_1 = \lambda_2 = k$.

=¿ if $\lambda_2 = k$, for $f \perp \mathbf{1}$ and $f \neq 0$ eigenvector of λ_2:

$$\langle f, (kI - A)f\rangle = \sum_{x\in V} f(x)\big((kI - A)f(x)\big) = \sum_{x\in V}\sum_{y\sim x}\big(f(x)^2 - f(x)f(y)\big)(*)$$

and because G is an oriented and $\langle f, (kI - A)f\rangle = k\|f\| - k\|f\| = 0$:

$$(*) = \frac{1}{2}\sum_{x\in V}\sum_{y\sim x}\big(f(x) - f(y)\big)^2 = \sum_{\{x,y\}\in E}\big(f(x) - f(y)\big)^2 = 0$$

Hence $f(x) = f(y)$ but if X is connected f is constant on V but $f \perp \mathbf{1}$ and $f \neq 0$ so impossible hence X is not connected.

For the third part, it is similar to the second part: ¡= For this, with $G = G_1 \sqcup G_2$ and $G_1 = I_1 \sqcup O_1$ bipartite component of G, we need to use the vector:

$$f(x) = \begin{cases} 1 & \text{if } x \in I_1 \\ -1 & \text{if } x \in O_1 \\ 0 & \text{else} \end{cases}$$

to get $A \cdot f(x) = -kf(x)$ =¿ And the other side is done by doing $\langle f, (kI + A)f\rangle$ to get $f(x) = -f(y)$ for $x \sim y$ to get that G contains a non empty bipartite graph as a connected component. □

Definition 1.1.8. Let $\varepsilon > 0$ and $k \geq 1$. A finite k-regular graph is said to be a (one-sided) ε-expander graph if one has $\lambda_2 \leq (1 - \varepsilon)k$ and a two-sided ε-expander if one also has $\lambda_n \geq -(1 - \varepsilon)k$.

Remark 1.1.9. The definition of one-sided expander is equivalent to the definition of expander.

Proof. Let $S \subseteq V$ s.t. $0 \leq |S| \leq \frac{|V|}{2}$ with $s = |S|$ and $n = |V|$ Let $f = \mathbb{1}_S - \frac{s}{n}\mathbf{1}$, $\langle f, \mathbf{1}\rangle = s - \frac{s}{n} * n = 0$ =¿ By introduicing the quotient of Rayleigh:

$$\lambda_2 = \max_{\substack{g\neq 0 \\ g\perp 1}} \frac{\langle A \cdot g, g\rangle}{\langle g, g\rangle} \text{ hence } \frac{\langle A \cdot f, f\rangle}{\langle f, f\rangle} \leq \lambda_2 \leq (1 - \varepsilon)k$$

Then

$$\frac{\langle (kI - A) \cdot f, f\rangle}{\langle f, f\rangle} \geq \varepsilon k$$

And by definition of f:

$$\langle (kI - A) \cdot f, f\rangle = \sum_{\{x,y\}\in E}\big(f(x) - f(y)\big)^2 = |\partial_E S|$$

Hence:

$$\frac{|\partial S|}{s\left(1-\frac{s}{n}\right)} \geq \varepsilon k \frac{|\partial S|}{|S|} \geq \varepsilon k\left(1-\frac{s}{n}\right) \text{ with } 1-\frac{s}{n} \geq \frac{1}{2}\frac{|\partial S|}{|S|} \geq \frac{\varepsilon k}{2}$$

¡= Here we just need to "go back up" and use the max to bring up λ_2. □

Remark 1.1.10. A consequence of 1.1.7 is that a graph G is one-sided if and only if G is connected and two-sided if and only if G is also non-bipartite

$(A^m)_{ij} = \#\{\text{walks of length m from i to j}\} Tr(A^m) = \#\{\text{closed walks of length m}\}$

We can see it easily with:

$$(A^{m+1})_{ij} = \sum_{l \in V} (A^m)_{il} A_{lj}$$

Theorem 1.1.11 (Trace formula)**.** *Let G be a k-regular simple graph (i.e. we do not authorize loop) with n vertices $n > k \geq 1$:*

1. $\sum_{i=1}^{n} \lambda_i = 0$
2. $\sum_{i=1}^{n} \lambda_i^2 = nk$
3. $\max(\lambda_2|, |\lambda_n|) \geq \sqrt{k} - o_{n\to+\infty}(1)$

Proof. (1) We just need to use that $\sum_{i=1}^{n} \lambda_i = Tr(A)$ but because G is a simple k-regular graph $\sum_{i=1}^{n} \lambda_i = Tr(A) = 0$ We can also see it by seeing the $Tr(A)$ is the number of closed walks of length 1 but this is impossible because there is no loop.

(2) Because G is a k-regular graph: $\sum_{i=1}^{n} \lambda_i^2 = Tr(A^2) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij} = \sum_{i=1}^{n} k = nk$ We can also see it by taking any of the n vertices and because they have k neighbors and $Tr(A^2)$ is the number of closed walks of length 2 it corresponds just to start from any points and then go to any of his neighbor and go back to his starting points.

(3) From the fact that $\lambda_1 = k$ with (2), we get that $\sum_{i=2}^{n} \lambda_i^2 = nk - k^2 = k(n-k)$. Then we can look for $i \geq 2$ at the "mean" value of $\lambda_i^2 \sim \frac{k(n-k)}{n-1}$ But because $\lambda_2 \geq \cdots \geq \lambda_n$:

$$\max(|\lambda_2|^2, |\lambda_n|^2) = \max_{1\leq i\leq n}(|\lambda_i|^2) \geq k\frac{n-k}{n-1} = k - k\frac{k-1}{n-1}$$

And with the square root:

$$\max(|\lambda_2|, |\lambda_n|) \geq \sqrt{k} - o_{n\to+\infty}(1)$$

because $k\frac{k-1}{n-1} \xrightarrow[n\to+\infty]{} 0$ □

Example 1.1.12 (Complete graph)**.** Let $G = (V, E)$ be the complete graph with n vertices, $(n-1)$-regular and adjacency matrix A:

$$A = B - I_n \text{ with } B = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix}$$

=¿
Let $f = \big(f(1), \ldots, f(n)\big)$ be like $\sum_{i=1}^{n} f(i) = 0 (f \perp 1)$ so then $f \cdot B = 0$
$A \cdot f = -f$ means -1 is an eigenvalue with eigenspace $\{f \in \mathbb{R}^n / f \cdot B = 0\} = 1^{\perp}$ so it's the kernel of a linear form of $\mathbb{R}^n$ hence of dimension $n-1$ and $\lambda_2 = \cdots = \lambda_n = -1$ ¡=
If f is an eigenvector of the eigenvalue -1 then $(B - I_n)f = -f \implies B \cdot f = 0 \implies \sum_{i=1}^{n} f(i) = 0$.

$$\begin{aligned} \lambda_2 &\leq (1-\varepsilon)k? \\ -1 \leq -\frac{1}{n-1} \times (n-1) &= -1 \\ \lambda_n &\geq -(1-\varepsilon)k? \\ -1 \geq -\Big(1 - 1 + \frac{1}{n-1}\Big) &\times (n-1) \\ -1 &\geq -1 \end{aligned}$$

Hence G is one-sided $\big(1 + \frac{1}{n-1}\big)$-expander and a two-sided $\big(1 - \frac{1}{n-1}\big)$-expander.

Example 1.1.13 (Bipartite complete graph)**.** Let $n \geq 2$ be an even number and let $G = K_{\frac{n}{2}, \frac{n}{2}}$ the complete bipartite graph between two sets of $\frac{n}{2}$ vertices each, thus G is a $\frac{n}{2}$-regular graph and $\lambda_n = -\frac{n}{2}$ and $\lambda_2, = \cdots = \lambda_{n-1} = 0$.

Theorem 1.1.14 (Poincaré inequality)**.** *Let $G = (V, E)$ be a k-regular graph and $f : V \longrightarrow \mathbb{C}$ a function on vertices. We define the gradient magnitude:*

$$\begin{aligned} |\nabla f| : V &\longrightarrow \mathbb{R}_{\geq 0} \\ v &\longmapsto |\nabla f|(v) = \left(\sum_{\substack{w \in V \\ \{v,w\} \in E}} |f(w) - f(v)|^2 \right)^{\frac{1}{2}} \end{aligned}$$

and $\|\nabla f\|^2_{l^2(V)} = \sum_{v \in V} \sum_{w \sim v} |f(w) - f(v)|^2$.
Then the Poincaré inequality *says that, G is a one-sided ε-expander if and only*

if:

$$\|\nabla f\|^2_{l^2(V)} \geq 2k\varepsilon\|f\|^2_{l^2(V)}$$

When ever f has mean 0 i.e. $\sum_{v\in V} f(v) = 0$

Proof.

$$\|\nabla f\|^2_{l^2(V)} = \sum_{v\in V}\sum_{w\sim v}|f(w)|^2 + \sum_{v\in V}\sum_{w\sim v}|f(v)|^2 - 2\sum_{v\in V}\sum_{w\sim v} Re\big(f(w)\overline{f(v)}\big)$$

We get by identifying the 2nd term with the equivalent relation $w \sim v$ to be equal to the 1st term and the 3rd term:

$$\|\nabla f\|^2_{l^2(V)} = 2k\|f\|^2_{l^2(V)} - 2\langle A\cdot f, f\rangle$$

f means $f \perp 1$ but on the orthogonal space the highest eignevalue is λ_2 so if $f = a_2v_2 + \cdots + a_nv_n$

$$\frac{\langle A\cdot f, f\rangle}{\|f\|^2_{l^2(V)}} = \frac{\lambda_2|a_2|^2 + \cdots + \lambda_n|a_n|^2}{|a_2|^2 + \cdots + |a_n|^2} \leq \lambda_2$$

$$\langle A\cdot f, f\rangle \leq \lambda_2\|f\|^2_{l^2(V)}$$

$$\|\nabla f\|^2_{l^2(V)} \geq (2k - 2\lambda_2)\|f\|^2_{l^2(V)}$$

=¿
If G is a one-sided ε-expander:

$$\|\nabla f\|^2_{l^2(V)} \geq 2k\varepsilon\|f\|^2_{l^2(V)}$$

¡=
Let f_2 be an eigenvertex associated to λ_2 as $f_2 \perp 1$ because of the mean zero and $A\cdot f_2 = \lambda_2 f_2$

$$\|\nabla f_2\|^2_{l^2(V)} = (2k - 2\lambda_2)\|f\|^2_{l^2(V)} \geq 2k\varepsilon\|f\|_{l^2(V)}$$

$$\implies k(1-\varepsilon) \geq \lambda_2$$

□

Proposition 1.1.15 (Cheeger inequalities)**.** *In previous sessions, we saw Cheeger inequalities for a graph G and we can "translate" it with eigenvalues as:*

$$\frac{k-\lambda_2}{2} \leq h(G) \leq \sqrt{(k+\lambda_2)(k-\lambda_2)}$$

Remark 1.1.16. The proof [2][3] is a bit long so I will just give the idea of it:

1. For the left inequality, it is really similar to the way we proved that one-sided expander and the other definition of expander are equivalent using Rayleigh quotient.

2. For the right inequality, it is a bit harder and longer but the idea of it is to order the vertices depending of the value of $g(v_i)^2$ for a positive non-zero g with $|\text{supp}(g)| \leq \frac{n}{2}$, take $S_i = \{v_1, \ldots, v_i\}$ and from this use Cauchy-Schwartz to get the result.

Another way to understand Cheeger constant is:

$h(X) = 0 \iff X$ is not connected

$h(X) \gg 1$ (large) $\iff$ highly connected, i.e., many edges must be removed to disconnect the graph.

In this way, I let this as an exercise:

Question 1.1.17. Let G be k-regular graph with n vertices that is a one-sided ε-expander for some $n > k \geq 1$ and $\varepsilon > 0$. Show that if one removes m edges from G for some $m \geq 0$ then the resulting graph has a connected component of size at least $n - Cm$ where C is a constant depending only on k and ε.

Definition 1.1.18 (prime closed geodesic)**.** On a graph $G = (V, E)$, we called g a *prime closed geodesic* or *reduced closed walk* means a non-backtracking closed walk (i.e. $v_{i-1} \neq v_{i+1}$) and *prime/primitive* means g can't be obtained by repeating a closed geodesic $\widetilde{g}$ m times for $m > 1$ integer

Definition 1.1.19 (Zeta function of Ihara)**.** We define the zeta function of Ihara on a k-regular graph $G = (V, E)$ with $k > 1$, for $U_k = \{z \in \mathbb{C}/|z| < \frac{1}{k-1}\}$ such that:

$$\zeta_G(u) = \prod_{p \text{ prime closed geodesic}} \frac{1}{1 - u^{l(p)}}$$

Where l is the word length.

Remark 1.1.20. We can extend this definition on any graph $G = (V, E)$ by defining the $2|E| \times 2|E|$ non-backtracking operator matrix B (also called Hashimoto matrix) index by elements of oriented edges $\overrightarrow{E}$ such that:

$$B_{(u,v),(w,x)} = \begin{cases} 1 & \text{if } v = w \text{ and } u \neq x \\ 0 & \text{else} \end{cases}$$

with (u, v) and (w, x) are oriented edges and $U_G = \{z \in \mathbb{C}/|z| < \frac{1}{\rho(B)}\}$

Theorem 1.1.21 (Determinant formula of Bass-Ihara (1992) [4][5])**.** *From this extension on a graph $G = (V, E)$, we define the determinant of Bass-Ihara like this:*

$$\frac{1}{\zeta_G(u)} = \det(I - uB) = (1 - u^2)^{|E|-|V|} \det(I - uA + u^2(D - I_n)$$

where $n = |V|$ and $D = Diag\big(\sum_{j=1}^{n} A_{1j}, \ldots, \sum_{j=1}^{n} A_{nj}\big)$

Remark 1.1.22. From this, we have that ζ_G as a meromorphic extension on $\mathbb{C}$.

Definition 1.1.23 (Trace formula of Ihara)**.** With the zeta function of Ihara, we also get a new trace formula for the Hashimoto matrix B by looking at:

$$\frac{\zeta_G'}{\zeta_G}(u) = \sum_{m\geq 1} Tr(B^m)u^{m-1} \text{ with } Tr(B^m) = \sum_{\substack{[p] \text{ prime closed geodesic} \\ l(p)|n}} l(p)$$

where $[p]$ is the equivalent class of prime closed geodesic where 2 geodesic are equivalent if they differ only by their starting points along the same oriented graph (e.g. on the complete graph $G = \{1,2,3\}$, $P_1 = \{1,2,3,1\}$, $P_2 = \{2,3,1,2\}$ and $P_3 = \{3,1,2,3\}$ are all equivalent).

From regular graph, we can see a lot of parallel with hyperbolic surface:

Regular graph	**Hyperbolic surface**
(primitive) reduced closed walk (i.e. $v_{i-1} \neq v_{i+1}$)	(primitive) closed geodesic
spectrum of $A/\Delta = kI - A$ (finite)	spectrum of Δ of Laplace-Beltrami (infinite discrete)
non-backtracking flow	geodesic flow
Trace formula/Ihara trace formula	Selberg trace formula
Determinant of Ihara-Bass	Determinant regulate of a spectral operator
Zeta function of Ihara	Zeta function of Selberg

1.2 Ramanujan graph, spectral gap and probability

Theorem 1.2.1 (Alon-Boppana (1991))**.** *Let $X_{n,k}$ be an infinite family of k-regular connected graphs on n vertices where k is fixed and $n \to +\infty$.*
Then $\lambda(X_{n,k}) = \max\{|\lambda|/|\lambda| \neq k$ and λ eigenvalue of A_n for all $n\} \geq 2\sqrt{k-1} - o(1)$

[6]. The idea is similar to the proof of the third result of the trace formula but with the power $2m$:

$$\mathrm{Tr}(A^{2m}) = k^{2m} + \sum_{i=2}^{n} \lambda_i^{2m} \leq k^{2m} + (n-1)\lambda(X_{n,k})^{2m}$$

$$\mathrm{Tr}(A^{2m}) \geq nC_m(k-1)^m, \quad \text{with } C_m = \frac{1}{m+1}\binom{2m}{m}$$

$$\liminf_{n\to+\infty} \lambda(X_{n,k}) \geq \sqrt{k-1}\, C_m^{\frac{1}{2m}}, \quad \text{with } C_m^{\frac{1}{2m}} \xrightarrow[m\to+\infty]{} 2.$$

We get the third inequality by using the first and second inequality and using the $2m$-square root and the $-o(1)$ is from the term $-\frac{k^{2m}}{n-1}$. □

Definition 1.2.2 (Ramanujan graph)**.** A k-regular finite graph X is called a *Ramnujan graph* if $\lambda(X) \leq 2\sqrt{k-1}$

Theorem 1.2.3 (Friedman (2008))**.** $\forall \varepsilon > 0$ *and* $k \geq 3$

$$\mathbb{P}(\lambda(X) \leq 2\sqrt{k-1} + \varepsilon) = 1 - o_k(1)$$

when X is a random (n,k)-graph (X is a random variable $X \sim Unif(\mathcal{G}_{n,k})$ a graph choosen randomly among the k-regular graph) i.e. almost every such X is almost Ramanujan.

Proof. The proof is very long so I will just give the idea of it:
$\mathbb{E}(Tr(A^{2m}) \simeq n(2\sqrt{k-1})^{2m} + \dots$
And use Markov inequality (but the hard part is the other terms).
There is different proof, the one in 2008 [7] use the trace formula in 100 pages, Bordenaire did it also in 2020 [8] in 25 pages, 2021 Huang-Yau [9] did it with random matrix and in 2024 Chen-Garza-Vargos-Tropp-Van Handel [10] in 30 pages but getting also strong convergence spectrum. □

What people would like to get is an existence of an infinite family of Ramanujan graphs to get an "optimized" infinite family with the theorem of Alon-Boppana. For $k = p + 1$ with p prime, it was already proved by Margulis-Lubotzy-Phill-Sarnak (1988) [11][12].

Theorem 1.2.4 (Marcus-Spielman-Srivastava [13])**.** $\forall k$, $\exists \infty$ *number of bipartite Ramanujan graphs*

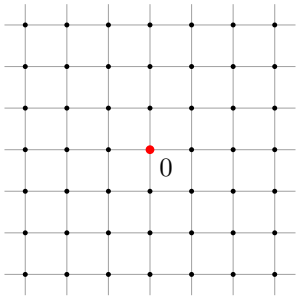
$\forall k$? it is still an open question.

2 Cayley graph

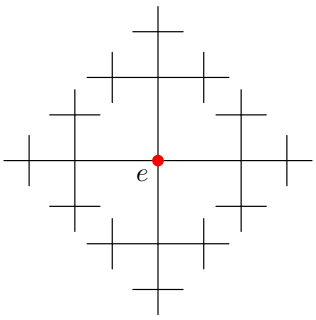
2.1 Construction and link with com surface

Definition 2.1.1 (Cayley graph)**.** Let Γ be a group generate by a finite set S. $Cay(\Gamma, S)$ is define by $V = \Gamma$ and we put edges between γ and $\gamma \cdot s \forall \gamma \in \Gamma$ and $s \in S \cup S^{-1}$ i.e. $\gamma \sim \widetilde{\gamma} \iff \widetilde{\gamma} = \gamma \cdot s$ for $s \in S \cup S^{-1}$

Example 2.1.2. The Cayley graph $\mathrm{Cay}(\mathbb{Z}^2, \{(1,0),(0,1)\})$ is the infinite square lattice.



Example 2.1.3. The Cayley graph of the free group $F_2 = \langle a, b \rangle$ with respect to the generating set $\{a, b\}$ is the infinite 4-regular tree.



Proposition 2.1.4 (Corollary of Švarc-Milnor)**.** *Let M be a compact riemannian manifold (non-empty), $\widetilde{M}$ universal cover,*

$$\begin{aligned} \pi_1(M) &\longrightarrow \widetilde{M} \\ g &\longmapsto g \cdot x \end{aligned}$$

This bring the useful result to see a better link between graph and hyperbolic surface and more generally with riemannian manifold:

$$Cay(\pi_1(M), S) \simeq_{Q.I.} \pi_1(M) \simeq_{Q.I.} \widetilde{M}$$

2.2 Random walk

Definition 2.2.1 (Random walk)**.** Let $G = (V, E)$ k-regular,
A *random walk* $(X_n)_{n \geq 0}$ is a sequence of random variables s.t.:

$$\mathbb{P}(X_{n+1} = y | X_n = x) = \begin{cases} \frac{1}{k} & \text{if} \\ x \sim y & \\ 0 & \text{else} \end{cases}$$

We say that a random walk is *recurrent* if it go back at his initial point in a finite time with probability 1 and else it is said to be *transient*.

2.3 Permutation representation, Schreirer graph and Kazhdan constant

Let Γ be a finitely generated group and let $S \subseteq \Gamma$ be a finite symmetric generating set, i.e. $S = S^{-1}$.

Definition 2.3.1 (Permutation representation)**.** Suppose that Γ acts on a finite set V.
This action induces a unitary representation

$$\begin{aligned} \pi : \Gamma &\longrightarrow U(\ell^2(V)), \\ g &\longmapsto \pi(g), \end{aligned}$$

where

$$(\pi(g)f)(v) = f(g^{-1} \cdot v).$$

This representation is called the *permutation representation* associated with the action of Γ on V.

Definition 2.3.2 (Schreier graph)**.** Let Γ act on a set V. The *Schreier graph* associated with this action and the generating set S, denoted by

$$X = \mathrm{Sch}(\Gamma \curvearrowright V, S),$$

is the graph whose vertex set is V, with an edge labelled by $s \in S$ joining v to $s \cdot v$, for every $v \in V$ and $s \in S$. Loops and multiple edges are allowed.

An important example arises when $H \leq \Gamma$ is a subgroup. The action of Γ on the left cosets Γ/H gives the Schreier graph

$$\mathrm{Sch}(\Gamma/H, S).$$

In particular, the Cayley graph is a special case:

$$\mathrm{Cay}(\Gamma, S) = \mathrm{Sch}(\Gamma/\{e\}, S).$$

Let $X = \mathrm{Sch}(\Gamma \curvearrowright V, S)$ and let $k = |S|$.
The adjacency operator of X can be written as

$$A = \sum_{s \in S} \pi(s).$$

Consequently, the graph Laplacian is

$$\Delta = kI - A = \sum_{s \in S} (I - \pi(s)).$$

Since S is symmetric, A is self-adjoint.

Definition 2.3.3 (Kazhdan constant)**.** Let $\pi : \Gamma \longrightarrow U(\mathcal{H})$ be a unitary representation on a Hilbert space $\mathcal{H}$.
The subspace of invariant vectors is

$$\mathcal{H}^\Gamma = \{v \in \mathcal{H} : \pi(g)v = v, \ \forall g \in \Gamma\}.$$

The *Kazhdan constant* of π with respect to S is defined by

$$\kappa(\Gamma, S, \pi) = \inf_{\substack{v \in (\mathcal{H}^\Gamma)^\perp \\ \|v\|=1}} \max_{s \in S} \|\pi(s)v - v\|.$$

The Kazhdan constant of the group Γ is

$$\kappa(\Gamma, S) = \inf_{\substack{\pi \text{ unitary} \\ \mathcal{H}^\Gamma = \{0\}}} \kappa(\Gamma, S, \pi).$$

A finitely generated group Γ has *Kazhdan's property (T)* if

$$\kappa(\Gamma, S) > 0.$$

Equivalently, there exists $\varepsilon > 0$ such that for every unitary representation π and every $v \in (\mathcal{H}^\Gamma)^\perp$,

$$\max_{s \in S} \|\pi(s)v - v\| \geq \varepsilon \|v\|.$$

Proposition 2.3.4 (Expander Schreier graph)**.** *Let $X = \mathrm{Sch}(\Gamma/H, S)$ be a finite Schreier graph, where S is symmetric and $|S| = k$.*
Suppose that the associated permutation representation π satisfies

$$\kappa(\Gamma, S, \pi) \geq \varepsilon > 0.$$

Then the second largest eigenvalue of the adjacency operator satisfies

$$\boxed{\lambda_2(X) \leq k - \frac{\varepsilon^2}{2}.}$$

[14]. We admit this result (the idea of the proof is to decompose the set $l^2(V) = \mathbb{C}1 \oplus l_0^2(V)$ where $l_0^2(V)$ is the subspace of function with mean 0 of $l^2(V)$ and to use Rayleigh quotient with λ_2). □

Let Γ be a finitely generated group and let S be a finite symmetric generating set with $|S| = k$.
Let $\mathcal{L} = \{N_i\}_{i \in I}$ be a family of finite-index normal subgroups of Γ.
For each $i \in I$, consider the finite Cayley graph

$$X_i = \mathrm{Cay}(\Gamma/N_i, \overline{S}_i),$$

where $\overline{S}_i$ denotes the image of S in Γ/N_i, counted with multiplicity.
Let A_i be the adjacency operator of X_i with eigenvalues

$$k = \lambda_1(X_i) \geq \lambda_2(X_i) \geq \cdots \geq \lambda_{|V_i|}(X_i).$$

Definition 2.3.5 (Property (τ))**.** We say that Γ has property (τ) with respect to $\mathcal{L}$ if there exists $\varepsilon > 0$ such that

$$\lambda_2(X_i) \leq k - \varepsilon \qquad \text{for every } i \in I.$$

In other words, the family $(X_i)_{i \in I}$ has a uniform spectral gap.

Equivalently, writing

$$\Delta_i = kI - A_i,$$

property (τ) means that the first nonzero eigenvalue of Δ_i is uniformly bounded away from zero.
More precisely, there exists $\varepsilon > 0$ such that

$$\langle \Delta_i f, f \rangle \geq \varepsilon \|f\|^2$$

for every $i \in I$ and every $f \in \ell_0^2(V_i)$.

Proposition 2.3.6 ([15]). *The group Γ has property (τ) with respect to $\mathcal{L}$ if and only if the family*

$$\left(\mathrm{Cay}(\Gamma/N_i, \overline{S}_i)\right)_{i\in I}$$

has a uniform spectral gap. Equivalently, these graphs form a uniformly expanding family.

Remark 2.3.7. Property (T) implies property (τ) with respect to every family of finite-index normal subgroups.

The main link can be summarized as

$$\boxed{\begin{array}{c}\text{Property (T)} \Longrightarrow \text{Property } (\tau)\\ \Longrightarrow \text{Uniform spectral gap}\\ \Longrightarrow \text{Expander families.}\end{array}}$$

3 Fun result

Definition 3.0.1 (Chromatic number)**.** Let $G = (V, E)$ be a graph,
$\chi(G)$ = smallest number of color needed to color each vertices s.t. all neighbors have different color

Example 3.0.2. For a bipartite graph, we need 2 colors and for a complete graph we need $|V|$ colors.

Definition 3.0.3 (Systole/girth)**.**

$$\left.\begin{array}{r} girth(G)\\ systole(G)\end{array}\right\} = \text{ length of the smallest closed path (cycle) non empty}$$

Example 3.0.4. $girth(Tree) = +\infty$

Question 3.0.5.

$$\forall L \geq 1, \exists C \geq 1 \text{ s.t. } girth(G) \geq L \implies \chi(G) \leq C?$$

The answer is no !

Theorem 3.0.6 (Erdos [16])**.**

$$\exists (G_n)_{n\geq 0} s.t. \begin{cases} systole(G_n) \xrightarrow[n\to+\infty]{} +\infty \\ \chi(G_n) \xrightarrow[n\to+\infty]{} +\infty \end{cases}$$

Definition 3.0.7 (Independence set and independence number)**.** An independence set of a graph $G = (V, E)$ is the set I of vertices s.t. $\forall u, v \in I, \{u, v\} \notin E$.

$$\mathrm{ind}(G) = \text{independence number} = \max_{I \text{ independent set}} |I|$$

Proposition 3.0.8. *[[16]] For a graph $G = (V, E)$,*

$$\chi(G) \geq \frac{|V|}{ind(G)}$$

Proposition 3.0.9. *Let $G = (V, E)$ be a k-regular two-sided ε-expander graph with $n = |V|$,*
Let $I \subseteq V$ independence set, then:

1. $|I| \leq (1 - \varepsilon)n$
2. $\chi(G) \geq \frac{1}{1-\varepsilon}$

Proof. (2) is just a consequence of (1) with the inequality from 3.0.8. For (1), it comes from another result called the "expander mixing lemma" using $F_1 = F_2 = I$:

$$\left| E(F_1, F_2) - \frac{k}{n}|F_1||F_2| \right| \leq (1 - \varepsilon)k\sqrt{|F_1||F_2|}$$

□

If you want there is some more exercises on the website of Terence Tao [17], you can also find more details of how to construct family of expander graph with the Zig-Zag construction and also some application to Number theory in [1] and if you are more interested with the application of spectral geometry on hyperbolic surfaces there is the other lectures of Nalini Anantharaman at the Collège de France on youtube [16].

4 References

[1] Alexander Lubotzky. *Expander Graphs in Pure and Applied Mathematics*. 2011. arXiv: 1105.2389 [math.CO]. URL: https://arxiv.org/abs/1105.2389.

[2] Noga Alon and Vitali D. Milman. "λ_1, Isoperimetric Inequalities for Graphs, and Superconcentrators". In: *Journal of Combinatorial Theory, Series B* 38.1 (1985), pp. 73–88. DOI: 10.1016/0095-8956(85)90092-9. URL: https://web.math.princeton.edu/~nalon/PDFS/Publications2/lambda%20%20isoperimetric%20inequalities%20for%20graphs%20and%20superconcentrators.pdf.

[3] Jozef Dodziuk. "Difference Equations, Isoperimetric Inequality and Transience of Certain Random Walks". In: *Transactions of the American Mathematical Society* 284.2 (1984), pp. 787–794. DOI: 10.1090/S0002-9947-1984-0743744-X.

[4] Yasutaka Ihara. "On Discrete Subgroups of the Two by Two Projective Linear Group over $\mathfrak{p}$-Adic Fields". In: *Journal of the Mathematical Society of Japan* 18.3 (1966), pp. 219–235. DOI: 10.2969/jmsj/01830219. URL: https://doi.org/10.2969/jmsj/01830219.

[5] Hyman Bass. "The Ihara–Selberg Zeta Function of a Tree Lattice". In: *International Journal of Mathematics* 3.6 (1992), pp. 717–797. DOI: 10.1142/S0129167X92000357.

[6] Yufei Zhao. *Graph Theory and Additive Combinatorics: Exploring Structure and Randomness*. MIT OpenCourseWare, course 18.225, Fall 2023, lecture notes. Cambridge University Press, 2023. URL: https://ocw.mit.edu/courses/18-225-graph-theory-and-additive-combinatorics-fall-2023/mit18_225_f23_lec_full.pdf.

[7] Joel Friedman. *A Proof of Alon's Second Eigenvalue Conjecture and Related Problems*. Vol. 195. Memoirs of the American Mathematical Society 910. American Mathematical Society, 2008. DOI: 10.1090/memo/0910. URL: https://doi.org/10.1090/memo/0910.

[8] Charles Bordenave. "A New Proof of Friedman's Second Eigenvalue Theorem and Its Extension to Random Lifts". In: *Annales Scientifiques de l'École Normale Supérieure* 53.6 (2020), pp. 1393–1439. DOI: 10.24033/asens.2450. arXiv: 1502.04482. URL: https://arxiv.org/abs/1502.04482.

[9] Jiaoyang Huang and Horng-Tzer Yau. "Spectrum of Random d-Regular Graphs Up to the Edge". In: *Communications on Pure and Applied Mathematics* (2023). DOI: 10.1002/cpa.22176. arXiv: 2102.00963. URL: https://arxiv.org/abs/2102.00963.

[10] Chi-Fang Chen et al. “A New Approach to Strong Convergence”. In: *Annals of Mathematics* 203.2 (2026), pp. 555–602. DOI: 10.4007/annals.2026.203.2.4. arXiv: 2405.16026. URL: https://arxiv.org/abs/2405.16026.

[11] Alexander Lubotzky, Ralph Phillips, and Peter Sarnak. “Ramanujan Graphs”. In: *Combinatorica* 8.3 (1988), pp. 261–277. DOI: 10.1007/BF02126799. URL: https://math1.math.huji.ac.il/~alexlub/PAPERS/ramanujan%20graphs/ramanujanGraphs.pdf.

[12] Gregory A. Margulis. “Explicit Group-Theoretical Constructions of Combinatorial Schemes and Their Application to the Design of Expanders and Concentrators”. In: *Problems of Information Transmission* 24.1 (1988). Russian original: Problemy Peredachi Informatsii 24(1), 51–60.

[13] Adam W. Marcus, Daniel A. Spielman, and Nikhil Srivastava. “Interlacing Families I: Bipartite Ramanujan Graphs of All Degrees”. In: *Annals of Mathematics* 182.1 (2015), pp. 307–325. DOI: 10.4007/annals.2015.182.1.7. arXiv: 1304.4132. URL: https://arxiv.org/abs/1304.4132.

[14] Emmanuel Breuillard. *Expander Graphs, Property τ and Approximate Groups*. PCMI lecture notes. URL: https://www.math.u-psud.fr/~breuilla/BreuillardPCMI.pdf.

[15] Alexander Lubotzky. *Discrete Groups, Expanding Graphs and Invariant Measures*. Vol. 125. Progress in Mathematics. Basel: Birkhäuser, 1994. ISBN: 9783764350758.

[16] Nalini Anantharaman. *Spectre des surfaces hyperboliques aléatoires (1) : Modèles de graphes réguliers et de surfaces hyperboliques aléatoires*. Cours de géométrie spectrale, année 2024–2025, vidéo publiée le 7 novembre 2024. Collège de France. Nov. 2024. URL: https://www.youtube.com/watch?v=aBqeKj3d-lM.

[17] Terence Tao. *254B: Expansion in Groups*. Lecture notes and blog posts, 2011–2012. 2011. URL: https://terrytao.wordpress.com/category/teaching/254b-expansion-in-groups/.

[18] Shlomo Hoory, Nathan Linial, and Avi Wigderson. “Expander Graphs and Their Applications”. In: *Bulletin of the American Mathematical Society* 43.4 (2006), pp. 439–561. DOI: 10.1090/S0273-0979-06-01126-8. URL: https://www.math.ias.edu/~avi/PUBLICATIONS/MYPAPERS/HLW06/hlw06.pdf.

[19] Clara Löh. *Geometric Group Theory: An Introduction*. Universitext. Chapter 3: Cayley Graphs. Springer, 2017. DOI: 10.1007/978-3-319-72254-2. URL: https://link.springer.com/book/10.1007/978-3-319-72254-2.

[20] Richard Grassl and Oscar Levin. *Exploring Combinatorial Mathematics*. 1st ed. Open textbook, first edition, Fall 2019. 2019. ISBN: 9781686579547. URL: https://openmathbooks.org/ecm/pdfs/ecm-tablet.pdf.